

Self-Consistent Field (SCF) Calculations Hands-on with Quantum ESPRESSO

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Quantum ESPRESSO Practical Tutorial

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A condensed, practical workshop: SCF, convergence, EOS, bands, DOS, metals, magnetism, relaxations, and graphene examples.

Outline

- 1 Hands-on: Visualization Tools
 - Introduction to XMGrace
 - Introduction to XCrySDen and VESTA
- 2 SCF Fundamentals
- 3 Key Approximations & Ingredients
- 4 Numerical Setup & Convergence
- 5 Practical Example: Silicon
- 6 Band Structure & DOS
- 7 Metals: Smearing
- 8 Magnetism (Spin-Polarized Runs)
- 9 Structural Optimization (relax) & Supercells
- 10 Graphene, Graphane, Graphene Oxide
- 11 Practical Tips & Workflows
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Overview: Day 1 Hands-on Session

What We'll Cover

Before diving into quantum calculations, we need to master the tools for:

- **Plotting data** — XMGrace for 2D graphs
- **Visualizing structures** — XCrySDen and VESTA for crystal structures

Important Notes

- File names are shown in **magenta**
- Commands to type are shown in **blue**
- Input parameters are shown in **dark green**

Useful Linux Commands

`vi/gedit/emacs` — open files `mkdir/mv/rm` — create/move/delete
`cat/less` — view files `grep` — search text

What is XMGrace?

XMGrace Overview

XMGrace is a plotting package for generating 2D graphs

- Export results in scalar and vector formats (publication-ready!)
- Simple function fitting capabilities
- Most tutorials use XMGrace or gnuplot

Getting Started

Launch XMGrace from terminal:

```
xmgrace
```

Exercise 1: Making a Graph from Data

Dataset: Volume-Energy Calculations

File: **example1.dat** contains Quantum ESPRESSO results

- Column 1: Volume
- Column 2: Total energy for FCC structure
- Column 3: Total energy for BCC structure

Step 1: View the data

```
vim example1.dat
```

Step 2: Launch XMGrace

```
xmgrace
```

Importing Data into XMGrace

- 1 In XMGrace menu: **Data** → **Import** → **ASCII**
- 2 Navigate to Day1/xmgrace directory
- 3 Click ONCE on **example1.dat**
- 4 In 'Load As' box, select **Block Data** and click OK
- 5 Select X data from **column 1**, Y data from **column 2**
- 6 Click **Apply** and close

Why Block Data?

Block Data allows us to choose specific columns to plot (we only want columns 1 and 2 first)

Customizing Axis Range

Step 1: Open Axis Properties

Plot → Axis Properties

For X Axis:

- Select **Edit: X Axis**
- Change X range: 20 to 45
- Click **Apply**

For Y Axis:

- Select **Edit: Y Axis**
- Change Y range: -220 to 0
- Click **Apply**

Adding Labels and Legends

X Axis Label:

- In Label String: Volume ($\text{\AA}^3$)
- This creates: Volume ($\text{\AA}^3$)
- Set Major Spacing: 4, Minor Ticks: 0

Y Axis Label:

- Label String: Energy (Hartree)

Add Legend:

- Double-click on plotted data
- In Legend box: type "FCC"
- Click **Apply**

Add Title:

- **Plot** → **Graph Appearance**, enter "FCC structure"

Improving Plot Appearance

Make axis labels clearer:

- **Plot** → **Axis Properties**
- Click **Axis label & bar** tab
- Font: Helvetica-Bold, Char size: 120
- Bar Width: 2

Adjust tick labels:

- Click **Tick labels** tab, Char size: 120
- Click **Tick marks** tab, Line width: 2

Add symbols and change line style:

- Double-click on curve
- Symbol Properties → Type: Circle
- Line Properties → Change Style and Color

Saving and Exporting

Save as XMGrace file (.agr):

- **File → Save As**
- Filename: **example1.agr**

Export as image (e.g., PNG):

- 1 **File → Print Setup**
- 2 Device: select PNG
- 3 **File → Print** (this creates **example1.png**)
- 4 **File → Exit**

Verify files created:

```
ls
```

XMGrace Tasks

Task 1

Practice with plotting options to reproduce a professional-looking graph

Task 2

Plot **both** Column 2 and Column 3 as functions of Column 1

- This will show FCC and BCC structures on the same plot
- Use different colors/symbols for each dataset
- Add appropriate legends

Hint

Import the data twice, selecting different Y columns each time

Exercise 2: Fitting Data

Objectives

- Fit dataset to nth order polynomial
- Fit dataset to custom equation

Fitting to nth Order:

- 1 Import column 1 (X) and column 2 (Y)
- 2 **Data → Transformations → Regression**
- 3 Select fit type (e.g., Quadratic)
- 4 Click **Accept**
- 5 A pop-up shows the equation fit

Task

Find a 2nd order fit to columns 1 and 3. Display raw data as symbols and fit as a line.

Visualizing Crystal Structures

Why Visualize?

- DFT input files: lattice parameters + atomic coordinates (numbers)
- Visualization: actual spatial arrangement (pictures!)
- Much better insight into structure

Two programs:

- **XCrySDen**: Directly reads Quantum ESPRESSO files
- **VESTA**: Better for high-quality figures (needs file conversion)

XCrySDen Basics

Launch XCrySDen:

```
xcrysdn
```

Load structure:

- File menu → Open [Day1/calculations/PbTiO3/pto.scf.in](#)
- Format: pw.x input format

Explore:

- Buttons at bottom: atom info, distances, angles
- **Exercise:** What is Ti coordination? Find Ti-O distances. Are they equal?

Export:

- **Tools** → **k-path** (for band structure calculations)
- File menu → Export to xsf format (for VESTA)

VESTA: High-Quality Visualization

Launch VESTA:

VESTA

Basic operations:

- Load xsf file from XCrySDen export
- Check distances/angles: buttons on left side
- Add bonds/polyhedra: **Edit** → **Bonds**
- Modify boundaries: **Objects** → **Boundaries**
- View unit cell info: **Edit** → **Edit data** → **Unit cell**
- Export image: **File** → **Export raster image** (PNG)

Visualizing Electron Density

Charge Density from Quantum ESPRESSO

Load cube file: [Day1/calculations/Si/si.rho.cube](#)

Customize visualization:

- Modify isosurface: **Objects** → **Properties** → **Isosurfaces**
- 2D plot: **Utilities** → **2D Data Display**

Exercise

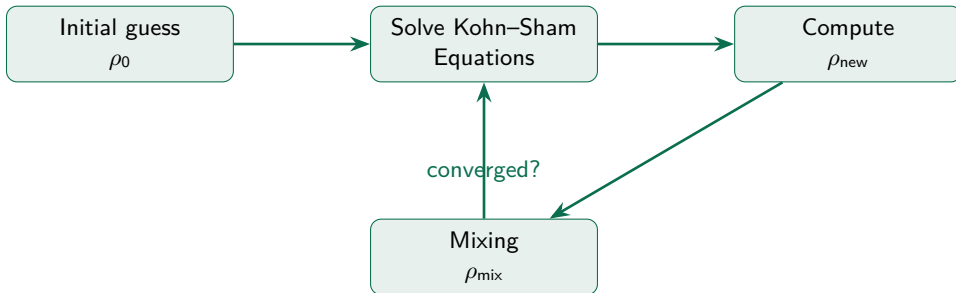
Pick a crystal structure from VESTA's library and analyze:

- What elements?
- Coordination polyhedra?
- How are they connected?

What is an SCF run? (High level)

- SCF = **Self-Consistent Field**. Iterative method solving the Kohn–Sham DFT equations.
- Goal: obtain ground-state electron density $\rho(\mathbf{r})$ and total energy E_{tot} .
- Loop: guess $\rho_0 \rightarrow$ build $V_{\text{KS}}[\rho] \rightarrow$ solve for $\{\psi_i\} \rightarrow$ get $\rho_1 \rightarrow$ mix $\rightarrow$ check convergence.

SCF loop (diagram)



Stop when $\|\rho_{n+1} - \rho_n\| < \epsilon$ (or $\Delta E < \text{tol}$).

Understanding the Convergence Criterion

Mathematical Expression

$$\|\rho_{n+1} - \rho_n\| < \varepsilon$$

- $\|\cdot\|$: Norm (magnitude/distance) — usually Euclidean distance
- ρ_n : Electron density at iteration n
- ρ_{n+1} : Electron density at next iteration $n + 1$
- ε (**epsilon**): Small tolerance value (e.g., 10^{-3} , 10^{-6})

Translation

Stop when the change between successive iterations is smaller than a threshold

Why SCF matters

- Provides the self-consistent electronic potential and eigenvalues — input to bands, DOS, response properties.
- SCF convergence quality affects all derived properties (energies, forces, phonons).
- Use conservative thresholds for production runs: $\text{conv_thr} \leq 10^{-8}$ (example).

Pseudopotentials: motivation and types

- **Core vs valence** — core electrons are chemically inert; valence electrons control bonding.
- **Pseudopotentials** (UPF files) replace nucleus + core electrons with a smooth effective potential.
- Types: Norm-conserving (NC), Ultrasoft (USPP), PAW. Choose according to accuracy/performance trade-offs.
- Example: `Si.pz-vbc.UPF` (LDA-type pseudo used in many examples).

Plane-Waves & Cutoffs: The Simple Story

The Core Idea (In Plain English)

Problem: We need to describe where electrons are in a crystal.

Solution: Use **waves** (like sound waves or ocean waves) to build electron shapes.

Think of it like building with LEGO blocks:

- **Plane waves** = Your LEGO blocks (simple building units)
- **Electron wavefunction** = The structure you build from those blocks
- **More blocks** = More detailed structures (but takes longer to build!)

The Key Question

How many LEGO blocks (plane waves) do we need?

→ **This is what the “cutoff” decides!**

Plane-wave basis & energy cutoffs

- Wavefunctions expanded in plane waves up to kinetic energy cutoff `ecutwfc`.
- Density cutoff `ecutrho` typically = 4–12× `ecutwfc` (USPP needs high `ecutrho`).
- Variational principle $\Rightarrow$ energy decreases monotonically with increasing basis size (cutoff).

$$E_{\text{cut}} = \frac{\hbar^2 G_{\text{max}}^2}{2m_e}$$

Understanding Plane-Wave Basis Sets

Why Plane Waves?

- **Plane waves:** $e^{i\mathbf{G}\cdot\mathbf{r}}$ where $\mathbf{G}$ are reciprocal lattice vectors
- Orthogonal, complete basis set — ideal for periodic systems (crystals)
- Wavefunctions: $\psi(\mathbf{r}) = \sum_{\mathbf{G}} c_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}}$

The Truncation Problem

Cannot include infinite plane waves — must truncate at some maximum $|\mathbf{G}|$

Energy Cutoff Explained

Cutoff Formula

$$E_{\text{cut}} = \frac{\hbar^2 G_{\text{max}}^2}{2m_e}$$

Breaking it down:

- $\hbar$: Reduced Planck constant ($h/2\pi$) — quantum scale
- G_{max} : Maximum reciprocal vector magnitude — highest frequency wave included
- m_e : Electron mass
- E_{cut} : Maximum kinetic energy of plane waves in the basis set

Physical Meaning

Only include plane waves with kinetic energy $\leq E_{\text{cut}}$. Higher cutoff = more waves = better accuracy (but slower!)

Why Two Different Cutoffs? (Simple Explanation)

The Photography Analogy

Think of taking a photo and then zooming in on details:

ecutwfc

The Original Photo

- Describes *one electron*
- Like a photo with certain resolution
- Set this value in input file

Example: `ecutwfc = 40`

ecutrho

The Zoomed Detail

- Describes *electron density* (many electrons)
- Needs **finer detail** (higher resolution)
- Usually 4–12× bigger!

Example: `ecutrho = 160`

Simple Rule

Density needs more detail than wavefunctions!

Because: $\text{density} = \text{wavefunction} \times \text{wavefunction}$ (multiplying creates finer details)

Two Different Cutoffs: `ecutwfc` vs `ecutrho`

`ecutwfc`

Wavefunction cutoff

- Controls expansion of $\psi(\mathbf{r})$
- Direct representation of electronic states
- Typical values: 30–100 Ry

`ecutrho`

Density cutoff

- Controls charge density $\rho = |\psi|^2$
- Product of two wavefunctions
- Needs higher frequencies!

Why `ecutrho` > `ecutwfc`?

- $\rho(\mathbf{r}) \propto \psi^*(\mathbf{r})\psi(\mathbf{r})$ — product doubles the frequency content
- **Norm-conserving PP**: `ecutrho` = 4× `ecutwfc` (automatic)
- **Ultrasoft PP (USPP)**: needs `ecutrho` = 8–12× `ecutwfc` (set manually!)

Variational Principle & Convergence

Variational Principle in Action

- Ground state energy is the **minimum** of the energy functional
- Larger basis set = more freedom to minimize energy
- $E_{\text{cut}}^{(1)} < E_{\text{cut}}^{(2)} \Rightarrow E_{\text{tot}}^{(1)} \geq E_{\text{tot}}^{(2)}$

Practical convergence strategy:

- 1 Start with low cutoff (e.g., 20 Ry)
- 2 Increase systematically: 30, 40, 50, 60 Ry...
- 3 Stop when $|\Delta E| < \text{tolerance}$ (e.g., 1 meV/atom)
- 4 Energy **decreases** (or stays same) — never increases!

Convergence Check

Plot E_{tot} vs ecutwfc — should see asymptotic approach to converged value

What are K-Points? (Simple Explanation)

The Big Picture

Problem: Crystals repeat infinitely — we can't calculate every atom!

Solution: Use **k-points** to sample the crystal's "recipe" instead of the whole crystal.

Think of it like taste-testing soup:

- You don't drink the whole pot to know if it tastes good
- You take a few **sample spoonfuls** from different locations
- More samples = better idea of the overall taste
- K-points are those "sample spoonfuls" in momentum space!

K-Point Grid Example

$4 \times 4 \times 4$ grid = 64 sample points in 3D space

More k-points = more accurate, but slower calculation

Brillouin Zone: Your Crystal's "Music Sheet"

Simple Analogy: Think of Music

Real space (crystal) = Time (when notes are played)

Brillouin Zone = Sheet music (which notes/frequencies exist)

What is it?

- A "map" of all electron wave patterns
- Like a box containing all possible wavelengths
- Every k-point = one specific wave pattern

Why k-points?

- Can't check every wave—too many!
- Sample a few representative ones
- Grid = evenly spaced samples

The Trade-off

Few k-points ($2 \times 2 \times 2$): Fast but rough → Quick tests

Many k-points ($12 \times 12 \times 12$): Slow but accurate → Final calculations

K-points and Brillouin-zone sampling

- Integrals over the BZ are replaced by sums over discrete **k-points**.
- Monkhorst–Pack grids: `K_POINTS automatic`
`nk1 nk2 nk3 k1 k2 k3`.
- Dense grids required for metals; convergence is not necessarily monotonic.

Convergence testing: recommended workflow

- 1 Fix pseudopotential and k-grid; sweep `ecutwfc` (16,20,24,28,32 Ry).
- 2 Fix cutoff; sweep k-point density ($2\times 2\times 2, 4\times 4\times 4, \dots$).
- 3 For metals, test smearing type and degauss.
- 4 For USPP, test `ecutrho` ratios: $\text{dual} = \text{ecutrho}/\text{ecutwfc} = 4, 6, 8, 12$.

Present results as tables and E vs parameter plots for clarity.

Silicon — quick facts

- Diamond structure (fcc with 2 atoms per cell).
- Use `ibrav=2`, `celldm(1)` (in a.u.), `nat=2`, `ntyp=1`.
- Typical test cutoffs: start 12–20 Ry; k-grid $6\times 6\times 6$ etc.

Silicon: SCF input (si.scf.in)

```
1 &CONTROL
2   calculation = 'scf',
3   prefix = 'silicon',
4   outdir = '../tmp/',
5   pseudo_dir = '../pseudo/',
6 /
7 &SYSTEM
8   ibrav = 2,
9   celldm(1) = 10.26,
10  nat = 2,
11  ntyp = 1,
12  ecutwfc = 12.0,
13 /
14 &ELECTRONS
15   conv_thr = 1.0d-8,
16   mixing_beta = 0.7,
17 /
18 ATOMIC_SPECIES
19 Si 28.086 Si.pz-vbc.UPF
20
21 ATOMIC_POSITIONS (crystal)
22 Si 0.00 0.00 0.00
23 Si 0.25 0.25 0.25
24
25 K_POINTS automatic
26 6 6 6 1 1 1
```